

Sources of fat

Storage of fats

Fat is stored in the adipose tissues. In normal human beings adipose tissue constitutes of 10 – 15% of the body weight.

Fat metabolism

Fatty acids are oxidized by certain enzymes in the tissues to carbon dioxide and water. The oxidation takes place through the tricarboxylic acid cycle.

Deficiencies

• Dry, scaly skin • Hair loss • Low body weight • Cold intolerance • Bruising	• Poor growth • Lower resistance to infection • Poor wound healing • Loss of menstruation

Over consumption

- Over weight
- Obesity
- Coronary heart disease
- Cancer
- High cholesterol

5.7 Proteins

Proteins are made of amino acids linked together by peptide bonds. Amino acids can be divided into essential amino acids and non – essential amino acids. Proteins and carbohydrates contain 4 kcal per gram.

Composition

Proteins are chemical compounds that contain the same atoms as carbohydrate and lipid – carbon (C), hydrogen (H) and oxygen (O) – but proteins are different in that they also contain nitrogen (N) atoms. These nitrogen atoms gives the name 'amino' (nitrogen containing) to the amino acids that are the links in the chains referred to as proteins.

Classification of proteins

Proteins are large molecules formed by the combination of a number of aminoacids. About 20 amino acids have been found to occur in proteins and are important from the point of view of human nutrition.

Table 4

Classification of Proteins (Based on chemical composition)				
Simple		**Conjugated**	**Derived**	
Globular proteins • Albumins • Globulins • Glutelins • Prolamines • Histones • Globins • Protamines	Scleroproteins • Collagens • Elastins • Keratins	• Nucleoproteins • Glycoproteins • Mucoproteins • Lipoproteins • Phoshoproteins • Chromoproteins • Matalioproteins	Primary • Coagulated Proteins • Proteans • Metaproteins	Secondary • Proteoses • Peptones • Polypeptides • Peptides

Unit_5_Applied_nutrition_Gen_Nur.indd 116 04-03-2020 17:12:18